SHAIL PRADHAN

Software Engineer

Mobile: +91-98321-47441 | Email: shail.pradhan@icloud.com |Location: India

EDUCATION

Sikkim Manipal Institute of Technology, Sikkim Manipal University
Bachelor of Technology (B. Tech.) in Computer Science and Engineering
Specialization: Artificial Intelligence and Machine Learning
Expected Graduation: 2026 | **CGPA:** 8.80

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning, Software Engineering, Object-Oriented Programming, Database Management Systems

WORK EXPERIENCE

NCompass TechStudio | Chennai, India
Frontend Engineer | Internship
September 2025 – December 2025

- Optimized large data tables by **implementing API-level pagination with loading states**, reducing initial page load time by approximately **30%** and enabling smooth interaction with high-volume datasets.
- Identified and resolved **UI and functional bugs** in a production React application, improving overall stability and user experience.
- Worked with **React.js, Redux, TypeScript, CSS, Rest APIs, Git, GitHub, and Material UI**, following team code review standards and version control best practices.

IT Department, Government of Sikkim
Full Stack Developer | Internship
May 2025 – July 2025

- Built an **end-to-end grievance automation system** using NLP, prompt engineering, semantic filtering, and automated routing, achieving approximately **90% classification accuracy**.
- Integrated the **Facebook Graph API** and developed a **real-time analytics dashboard**, enabling automated complaint capture, AI-driven prioritization, and improved operational visibility for faster decision-making.
- Worked across the stack using **JavaScript, Python, Django, MongoDB, CSS, and Postman**, following team collaboration practices and participating in code reviews.

PROJECTS

Agentic Rag Chatbot | (Next.js, TypeScript, Postgres, AWS S3, Vector database)

- **Cut context size by 7 times the original** using **Lang Chain** for PDF compression and semantic chunking strategies, achieving optimized token density that fits strict LLM context windows.
- Developing granular data fetching strategies. **Accomplished a 2-s reduction in Initial Content Load time by eliminating redundant API calls** and rendering only viewport-necessary data.
- Employed a custom document compressor abstraction to execute query-aware filtration and **accomplished a strategic separation of recall and precision that fetches 2x more initial candidates** while filtering out semantic noise to minimize LLM token usage.

Autonomous Navigation Agent using Deep Reinforcement Learning in Unity

- Contributed to a Unity ML-Agents **autonomous navigation project** using **Deep Reinforcement Learning (PPO)** with a **Python backend** and **GPU acceleration (CUDA, cuDNN)** for training..
- Applied simulation-based learning, policy optimization, and environment interaction workflows.

SKILLS

C/C++, Java, Python, TypeScript, JavaScript, SQL, Git, GitHub, Postman, TensorFlow, REST API, AWS, EC2, S3, Lambda, Kubernetes, Docker, Express, Redis, Next, React, Redux, MongoDB, PostgreSQL, MySQL

- Engineering: DSA, OOP, Complexity Analysis
- Mindset: Learning Agility, Problem Decomposition, Ownership

Core Engineering Competencies:

Product Development (E2E), Application Development, Object-Oriented Design, Data Structures & Algorithms, Time and Space Complexity Analysis, Multi-language Programming (C/C++, Java, Python, JS/TS)

AWARDS/ CERTIFICATES

- Top 0.05% ranked team on Adobe India Hackathon 2025.
- LeetCode 100+ problems solved. CodeChef rating: 1190